HW5

P5.4 a.) $u_x = \sin\theta$ $\quad \phi = 0$ $\quad$ Prove $\cos\theta = \pm \frac{n_x}{n_y}\sqrt{\frac{n_z^2 - n_y^2}{n_z^2 - n_x^2}}$

$u_y = 0$

$u_z = \cos\theta$

$$\frac{u_x^2}{n^2 - n_x^2} + \frac{u_y^2}{n^2 - n_y^2} + \frac{u_z^2}{n^2 - n_z^2} = \frac{1}{n^2}$$

$$\hookrightarrow u_x^2\left(\frac{1}{n^2} - \frac{1}{n_y^2}\right)\left(\frac{1}{n^2} - \frac{1}{n_z^2}\right) + \cancel{u_y^2\left(\frac{1}{n^2} - \frac{1}{n_x^2}\right)\left(\frac{1}{n^2} - \frac{1}{n_z^2}\right)}^{0} + u_z^2\left(\frac{1}{n^2} - \frac{1}{n_x^2}\right)\left(\frac{1}{n^2} - \frac{1}{n_y^2}\right) = \frac{1}{n^2}$$

$$\hookrightarrow \sin^2\theta\left(\frac{1}{n^2} - \frac{1}{n_y^2}\right)\left(\frac{1}{n^2} - \frac{1}{n_z^2}\right) + \cos^2\theta\left(\frac{1}{n^2} - \frac{1}{n_x^2}\right)\left(\frac{1}{n^2} - \frac{1}{n_y^2}\right) = \frac{1}{n^2}$$

$$\left(\frac{1}{n^2} - \frac{1}{n_y^2}\right)\left[\sin^2\theta\left(\frac{1}{n^2} - \frac{1}{n_z^2}\right) + \cos^2\theta\left(\frac{1}{n^2} - \frac{1}{n_x^2}\right)\right] = 0$$

$$\frac{1}{n^2} - \frac{1}{n_y} = 0 \Rightarrow n = n_y$$

$$\sin^2\theta\left(\frac{1}{n^2} - \frac{1}{n_z^2}\right) = -\cos^2\theta\left(\frac{1}{n^2} - \frac{1}{n_x^2}\right)$$

$$\hookrightarrow \frac{1}{n_y^2} = \frac{1 - \cos^2\theta}{n_z^2} + \frac{\cos^2\theta}{n_x^2}$$

$$\frac{\sin^2\theta}{n^2} + \frac{\cos^2\theta}{n^2} = \frac{\sin^2\theta}{n_z^2} + \frac{\cos^2\theta}{n_x^2}$$

$$\hookrightarrow \frac{1}{n_y^2} - \frac{1}{n_z^2} = \cos^2\theta\left(\frac{1}{n_x^2} - \frac{1}{n_z^2}\right)$$

$$\frac{1}{n^2} = \frac{\sin^2\theta}{n_z^2} + \frac{\cos^2\theta}{n_x^2} \Rightarrow \frac{1}{n_y^2} = \frac{\sin^2\theta}{n_z^2} + \frac{\cos^2\theta}{n_x^2}$$

$$\hookrightarrow \frac{n_z^2 - n_y^2}{n_y^2 n_z^2} = \cos^2\theta\left(\frac{n_z^2 - n_x^2}{n_x^2 n_z^2}\right) \Rightarrow \cos^2\theta = \frac{n_x^2(n_z^2 - n_y^2)}{n_y^2(n_z^2 - n_x^2)}$$

$$\Rightarrow \cos\theta = \pm\frac{n_x}{n_y}\sqrt{\frac{(n_z^2 - n_y^2)}{(n_z^2 - n_x^2)}}$$

b.) if $n_x \geq n_y \geq n_z$ $\quad \cos\theta = \pm\frac{n_x}{n_y}\sqrt{\frac{n_z^2 - n_y^2}{n_z^2 - n_x^2}} \Rightarrow 1 \leq \frac{n_x^2(n_z^2 - n_y^2)}{n_y^2(n_z^2 - n_x^2)}$

$$\Rightarrow n_y^2 n_z^2 - n_y^2 n_x^2 \leq n_x^2 n_z^2 - n_x^2 n_y^2 \Rightarrow n_y \leq n_x$$

so $n_x > n_z$ then $n_x \geq n_y \geq n_z$

if $n_z \geq n_y \geq n_x$ $\quad \cos^2\theta_1 = \pm\frac{n_x^2}{n_y^2}\left(\frac{n_z^2 - n_y^2}{n_z^2 - n_x^2}\right) = \sin^2\theta_2$

$$\underline{\theta_2 = \frac{\pi}{2} - \theta_1}$$

P5.6

$$n = n_e(\theta) = \frac{n_o n_e}{\sqrt{n_o^2 \sin^2\theta + n_e^2 \cos^2\theta}} \qquad \sin\theta_i = n_e(t') \sin\theta_t$$

$$\Rightarrow \sin\theta_i = \frac{n_o n_e}{\sqrt{n_o^2 \sin^2\theta_t + n_e^2 \cos^2\theta_t}} \sin\theta_t \Rightarrow \frac{1}{n_o^2 n_e^2} \sin^2\theta_i = \frac{1}{n_o^2 + n_e^2 \tan^2\theta_t}$$

$$\Rightarrow \frac{n_o^2 n_e^2}{\sin^2\theta_i} - n_o^2 = n_e^2 \tan^2\theta_t \Rightarrow \tan^2\theta_t = \frac{n_o^2}{\sin^2\theta_i} - \frac{n_o^2}{n_e^2}$$

$$\Rightarrow \frac{n_o^2 n_e^2}{n_e^2 \sin^2\theta_i} - \frac{n_o^2 \sin^2\theta_i}{n_e^2 \sin^2\theta_i} \Rightarrow \tan\theta_t = \frac{n_e \sin\theta_i}{n_o \sqrt{n_e^2 - \sin^2\theta_i}}$$